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Sequential Training of GANs Against GAN-Classifiers Reveals Correlated Knowledge Gaps

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AGENDA

- WHAT THE PAPER IS ABOUT
- MOTIVATION
- METHOD OVERVIEW
- KEY RESULTS
- RELATED WORK
- RESEARCH GAP
- PROPOSED HYPOTHESIS SOLUTION
- CONCLUSION

WHAT IS THE PAPER ABOUT

- Detection of GAN-generated images using classifiers
- Existence of consistent artifacts in GAN outputs
- Artifacts are shared across independently trained GANs
- Study interaction between GANs and external GAN-classifiers

MOTIVATION

- GAN images look realistic but still contain hidden artifacts
- External classifiers can detect GAN images reliably
- Indicates systematic “knowledge gaps” in GAN training
- Need to understand why these artifacts persist
- Need to study how GANs adapt when forced to remove artifacts

BACKGROUND (GAN OVERVIEW)

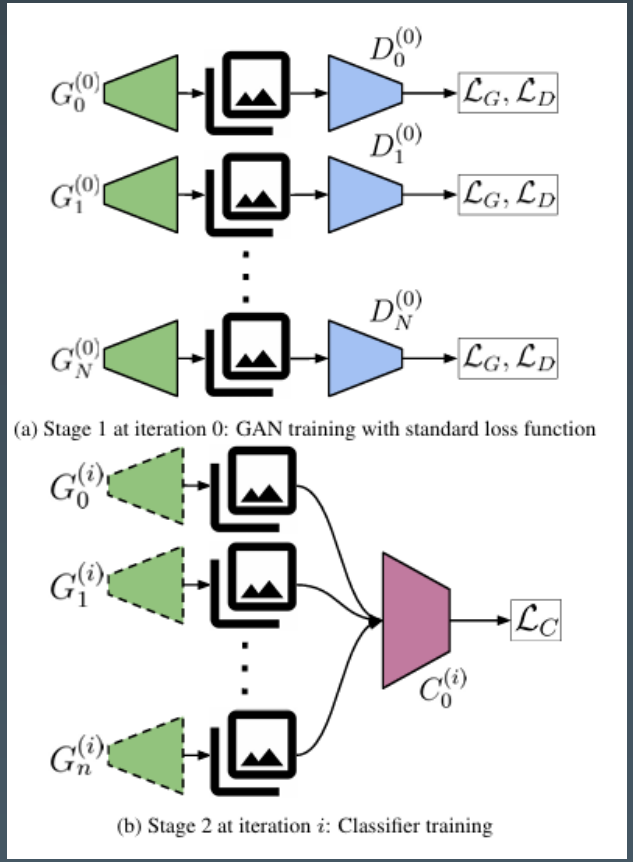
- Generator creates fake images from random noise
- Discriminator distinguishes real vs fake images
- Adversarial training between Generator and Discriminator
- Goal: Generator fools the Discriminator

PROPOSED METHOD

- Sequential training of GANs and GAN-classifiers
- Multiple GANs trained independently per iteration
- Classifiers trained on outputs from multiple GANs
- Modified GAN loss to also fool frozen classifiers
- Two strategies:
  - Fool-all loss
  - Memoryless loss

MODEL ARCHITECTURE DIAGRAM

- Generator with standard GAN discriminator
- Additional frozen classifier inputs to generator loss
- Iterative training across multiple rounds
- Separate classifier training stage



DATASET AND EXPERIMENTAL SETUP

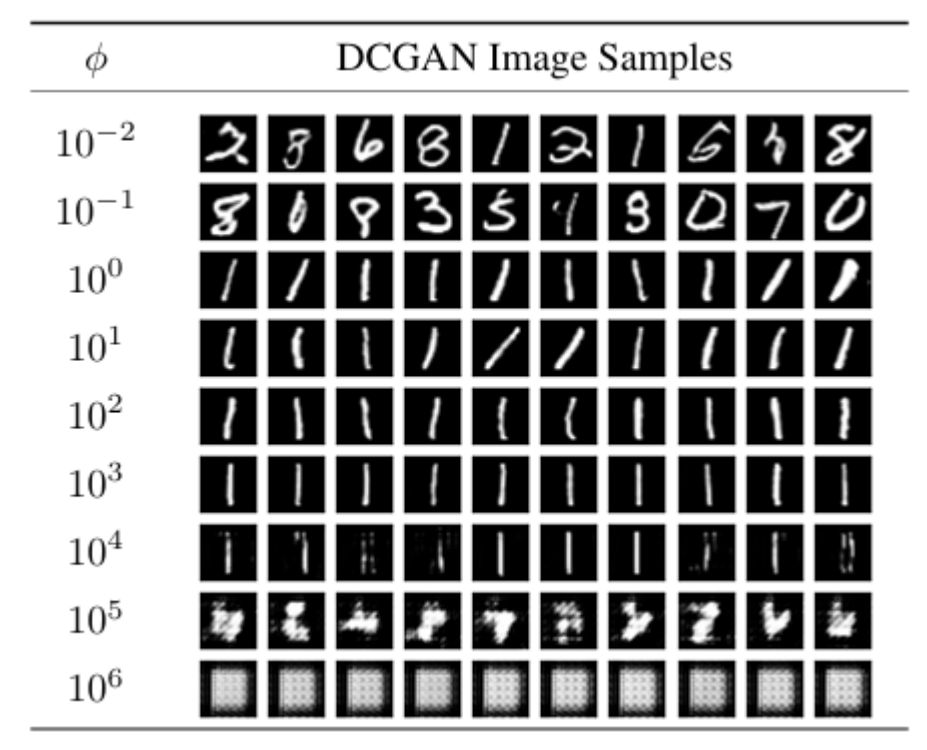
- MNIST dataset for low-dimensional setting
- FFHQ dataset for high-dimensional setting
- DCGAN used for MNIST
- StyleGAN2 used for FFHQ
- Classifier architectures:
  - ResNet-50
  - Inception-v3
  - MobileNetV2

RESULT & EVALUATION

- Evaluation metric: classification accuracy
- GAN fooling measured by drop in classifier accuracy
- FID used to measure image quality (StyleGAN2)
- Visual inspection for output quality
- Generalization tested on unseen GAN instances

VISUAL RESULTS

- GAN images maintain visual quality after training
- No degradation for StyleGAN2 across iterations
- DCGAN outputs degrade when classifier pressure increases



KEY FINDINGS

- GAN artifacts are correlated across generators
- DCGAN cannot fool classifiers without quality loss
- StyleGAN2 fool classifiers without quality loss
- Classifier capacity strongly affects learned artifacts
- Artifacts shift in an ordered, repeatable way

RELATED WORK

- CNN-Generated Images Are Surprisingly Easy to Spot (Wang et al., 2020)
- CNN-based GAN image detection
- Shows generalization across GAN architectures
- Does not study GAN adaptation to classifiers

RELATED WORK

- What Makes Fake Images Detectable? (Chai et al., 2020)
- Studies properties of detectable GAN artifacts
- Focuses on detection, not generator retraining
- No iterative GAN–classifier interaction

LIMITATION OF THE PAPER

- Limited to DCGAN and StyleGAN2 architectures
- Only image domain studied
- No semantic analysis of artifact types
- Computationally expensive experiments
- No real-world deployment evaluation

RESEARCH GAP

- No explicit localization of artifact regions
- No measurement of artifact semantic meaning
- Limited study of cross-architecture artifact overlap
- No adaptive or dynamic classifier selection
- Important for robust GAN detection and defense

PROPOSED SOLUTION

- Problem: GANs shift artifacts instead of removing them
- Hypothesis: Artifact localization improves artifact removal
- Introduce artifact-aware attention loss in GAN training
- Use classifier feature maps to guide generator updates
- Extend existing sequential GAN-classifier framework

CONCLUSION

- Paper reveals structured and correlated GAN artifacts
- Shows limits of GAN self-correction
- Highlight role of classifier capacity
- Proposed extension targets artifact localization
- Contributes toward more robust GAN generation

THANK YOU